

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Nature of roots

1. The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ **CBSE 2024, 1 mark**

SOLUTION

For real and equal roots, $D = b^2 - 4ac = 0$.

So $b^2 = 4ac$, that is $ac = \frac{b^2}{4}$.

Answer: (c)

2 Word problems

2. A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x . (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula. **CBSE 2024, 4 marks, case study**

SOLUTION

(i) The floor area is the same in both cases: $200x^2 = 128(x + 1)^2$

(ii) $200x^2 = 128x^2 + 256x + 128$, so $72x^2 - 256x - 128 = 0$. Dividing by 8: $9x^2 - 32x - 16 = 0$

(iii) By factorisation: $9x^2 - 36x + 4x - 16 = 0$, so $9x(x - 4) + 4(x - 4) = 0$ and $(9x + 4)(x - 4) = 0$

$x = 4$ or $x = -\frac{4}{9}$. A length cannot be negative, so $x = 4$.

OR by the formula: $x = \frac{32 \pm \sqrt{1024 + 576}}{18} = \frac{32 \pm 40}{18}$, which gives $x = 4$ or $x = -\frac{4}{9}$.

Answer: the side of a tile is 4 units.

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